
When Can Language Models Join Without Relation Names? Unique-Path Traversal versus End-to-End Accuracy under Path Ambiguity

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<https://github.com/pankajnits/oir>

Abstract

We ask whether 2-hop answering over small labeled graphs needs readable relation names, or only equality among identifiers. Opaque isomorphic reasoning (OIR) replaces relation strings with keyed opaque tokens, withholds a legend, and scores the answer node rather than an emitted (r_1, r_2) . A named-arm prompt names the condition in a one-line ARM header and uses readable case ids; hashed-id packing keeps that ARM line and hashes the ids; no-ARM packing drops the header. Unique sealed routes—one 2-hop from a start already in the question—still execute (OpenAI unique English/opaque 200/200; hashed-id no-ARM 200/200 at 512 tokens; opaque unique 32/32; both-opaque unique 32/32 on three families). When two complete same-type 2-hops compete and no hop is written, a CONTEXT-only instruction makes UNKNOWN the appropriate reply: CONTEXT gives no explicit cue for one city. Named-arm OpenAI usually yields that reply, not the decoy: 38/200 gold, UNKNOWN 151, decoy 11. English two-path is 199/200 under the same instruction, so the format does not force abstention when names identify a route. Packaging reduces abstention on those same two-path graphs, not a packaging-free opacity law (UNKNOWN 151 $\rightarrow$ 44 with hashed case ids, ARM line kept, gold 134/200, decoy 20; UNKNOWN 5 with the header dropped at 4096/medium). Dropping the header, at 4096 tokens and reasoning effort medium, is a two-path cell (unique-path was not run at that budget): cyclic-decoy English two-path is 197/200 and opaque two-path is 174/200. Entity names stay English and we score the answer node, not (r_1, r_2) , so two-path exact-match cannot be attributed to a join over CONTEXT. English relation names restore OpenAI with anonymous entities (opaque-entity two-path 32/32 vs. both-opaque 1/32). HMAC-SHA256 is one generator of that renaming on each instance’s finite atom set, not a cryptographic or privacy primitive. On the primary 2×2 , seals hide relation strings; entity names remain. They are not a confidentiality proof.

1 Introduction

The scored Wikidata question is *What city is the headquarters of the company led by Michael Eisner?* English unique-path context walks `works_at` then `headquartered_in` to Burbank. That success may use English predicates, or only equality among identifiers. The two-path decoy is an injected second person $\rightarrow$ company $\rightarrow$ city route (`partner_of/located_in`) to Kansas City, not a Wikidata fact. Opaque two-path keeps the English question and entity names; those predicates become keyed opaque tokens (`works_at` $\mapsto$ E334675310f05 on the scored Eisner item).

Opaque isomorphic reasoning (OIR) measures whether a model can join after English predicate strings are replaced by keyed opaque tokens, with no legend in the prompt. HMAC-SHA256 is the generator used here. A 2-hop factors as two inferred maps; we score only the answer node: selection $q \mapsto (r_1, r_2)$ and traversal $(v_0, r_1, r_2, G) \mapsto v_2$. Opacity does not, by itself, block traversal. The question is when the prompt no longer maps query semantics onto one of two sealed relation sequences. Putting the start entity in q as the same atom as in V (**start in q**) separates that question from start-entity identification. On the primary 2×2 , “lexical cues” means relation strings (entities stay English); both-opaque is a separate ablation. That suite is WIKI-H5. Prompts differ in how they name the condition: a readable header and case id (named-arm), the same header with hashed ids, or no header (Section 4).

This paper reports three measurements. (i) Unique sealed routes still execute with the start in q and no written hop. (ii) Under two competing sealed routes, named-arm OpenAI usually answers UNKNOWN, not the decoy. Under a CONTEXT-only instruction that is the appropriate reply: two same-type 2-hops are in CONTEXT and no hop is written, so CONTEXT gives no explicit cue for one city (Section 4). (iii) Matched demonstrations restore shown seals; novel seals are a negative control whose expected gold is near zero. Packaging moves the abstention rate on the same two-path graphs (Figure 1; Section 5.2). That packing result, including a two-path no-ARM English/opaque pair at 4096/medium, is not an opacity $\times$ topology law: unique-path was not run at that budget. What is new is named-arm abstention when opacity meets path ambiguity (Section 5.1), and how large a prompt-surface move can be on the same sealed graph (Section 5.2).

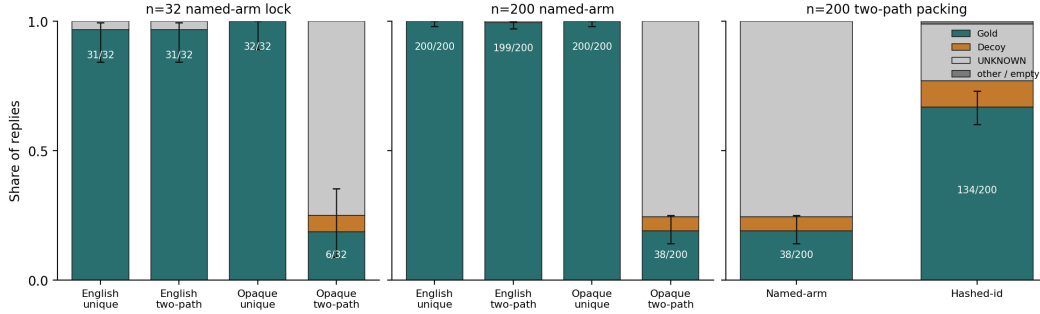


Figure 1: OpenAI WIKI-H5 (start in q). Stacked bars are gold / decoy / UNKNOWN; whiskers are Wilson 95% intervals. The usual named-arm reply is UNKNOWN, not the decoy. Left: $n=32$ named-arm lock (August; opaque two-path UNKNOWN 24, decoy 2, gold 6/32). Middle: city-typed $n=200$ named-arm (UNKNOWN 151, decoy 11, gold 38/200). Right: same $n=200$ graphs and ARM line, hashed case ids (packing on the same two-path graphs, not the named-arm headline) (gold 134/200, UNKNOWN 44, decoy 20; remainder other 1, NO_OUTPUT 1). No-ARM two-path at 4096/medium is Table 4. Alias $k=3$ is Table 3.

2 Related Work

Think-on-Graph and Reasoning-on-Graphs select the next readable relation (Sun et al., 2024; Luo et al., 2024). Ciphred ICL is not path ambiguity on a labeled graph (Fang et al., 2025; Guo et al., 2026). GSM-Symbolic swaps entity names in English math and templates some numerical values (Mirzadeh et al., 2025); 2WikiMultihopQA is a readable multi-hop control (Ho et al., 2020).

ARoG anonymizes *entities* while relations stay the retrieval language (Ning et al., 2026). PrivGemo HMAC-seals entities (ϕ_Q , e.g. ent_17); relations stay original labels or clustered schema words, not keyed opaque tokens, and a local Hand answers on the raw graph after a remote Brain ranks retrieved paths (CWQ 67% plaintext $\rightarrow$ 59% fully anonymized; WebQSP 75% fully anonymized) (Tan et al., 2026; Talmor and Berant, 2018; Yih et al., 2016). CoR explores readable relations because entities often have no semantics beyond opaque IDs — CoR’s opaque-entity hide-set, not relation-only WIKI-H5 (Liu et al., 2026). Tang et al. (2023) find LLMs fail when relation names are replaced by symbols ($\tau 1$) and rules must carry deduction; OpenAI unique-path opaque 32/32 is the complementary case: sealed relations still traverse when topology fixes the route. Guo et al. (2025, section 2.2) report collapse to random guessing among end entities when distractor two-hops have other sources.

Work	Hidden	Remote task	Join under opacity
ICL	substitution text	ICL task accuracy (bijection inferred)	ICL tasks
CIPHERS			
ARoG	entities; relations readable; topic names in q	retrieve + generate (relations, concepts)	privacy KG-RAG
PrivGemo	entities (HMAC); relations original or clustered schema words	rank retrieved paths; local plaintext answer	privacy KG-RAG
CoR	opaque entity IDs; relations readable	relation-centric exploration	KGQA
OIR	relations (primary 2×2); entities too (ablation); one join or none	emit graph-namespaces answer (no local decode)	selection vs. traversal

Table 1: Related protocols. PrivGemo HMAC-seals entities; relations remain original or clustered schema labels, and a local Hand answers on retrieved paths (CWQ/WebQSP). OIR withholds that ranking and local decode on the free arms and scores the answer token from the same model; named-arm two-path exact-match is not a CWQ rate. Both 2-hops still appear in CONTEXT.

Named-arm Wiki-H5 opaque two-path, with UNKNOWN allowed, is abstention rather than guessing (Section 5.1, Table 3). A forced-choice arm is not in this evaluation (Section 7) and would put those protocols on the same scale. OIR withholds a retrieved-path ranking and a local plaintext decoder on the free arms (Table 1); both 2-hops still appear in CONTEXT.

Chain-of-thought and Text-to-SQL externalize plans over English (Wei et al., 2022; Yu et al., 2018). We ask a behavioral question: can a pretrained model join when the relation vocabulary is freshly randomized and no plan is given? A sealed plan is then an execution upper bound. NLGraph and Talk like a Graph encode graphs without readable typed predicates (Wang et al., 2023; Fatemi et al., 2024); OIR starts from a labeled KG and removes only the relation lexicon. Case-id strings that name the condition (OPAQUE_AMBIG) are semantic content, not only a meaning-preserving format perturbation in the sense of Sclar et al. (2024); we still treat packing as a factor. Symbol tuning, binding-ID analyses, and compositional-limit studies change the labels or the task (Wei et al., 2023; Feng and Steinhardt, 2024; Dziri et al., 2023). OIR holds the join, hop length, triple count, and question template fixed and replaces relation strings; OpenAI unique-path opaque traversal stays at 32/32, so that removal withdraws the cue that would otherwise select between routes rather than blocking execution.

3 Problem Formulation

Let $G = (V, R, T)$ be a labeled multi-relational graph ($T \subseteq V \times R \times V$) and q a natural-language or programmatic query with answer $a \in V$. A **seal** σ is a keyed map from entity and relation strings to opaque tokens that we verify to be injective on the finite normalized atom set of each evaluation instance. We reject any instance where two distinct raw atoms normalize to the same string. Truncated HMAC-SHA256 (Krawczyk et al., 1997) is one generator of σ , not a cryptographic bound (Appendix A). The model receives G or $\sigma(G)$ and a query form built from σ (or from English, depending on the arm) and must return the answer node in that namespace; recovering σ^{-1} is not scored.

A 2-hop join factors as two maps we measure separately:

$$\text{selection: } (q, G, \mathcal{C}) \mapsto (r_1, r_2), \quad (1)$$

$$\text{traversal: } (v_0, r_1, r_2, G) \mapsto v_2. \quad (2)$$

Traversal is set-valued ($\tau_G: V \times R \times R \rightarrow 2^V$). We evaluate only instances for which $|\tau_G(v_0, r_1, r_2)| = 1$ for the gold plan (and, on two-path items, for the decoy plan), so each scored route has a unique tail. Uniqueness is per plan, not “one answer in G .” A sealed join plan supplies (r_1, r_2) and tests (2). The CEO free-English arm hashes q piecewise, so v_0 typically does not appear as a single graph atom; it is not a pure test of (1).

A 2-hop is *structurally executable* when (r_1, r_2) is given, or when topology leaves only one 2-hop from a start already in q . The prompt *explicitly maps* query semantics onto one relation sequence when it contains English predicates, a legend, or demonstrated seals. When two complete routes exist and relation tokens are arbitrary HMAC strings with no such mapping, selection is *unsupported by explicit cues* under this construction. Residual statistical structure can remain (token form, order, HMAC vs. permutation rates); we do not claim mathematical unidentifiability. Any gold on that arm must use residual prompt cues, parametric memory of the named entities, or other unmeasured information; we do not observe (r_1, r_2) . Low exact-match is not evidence that traversal failed. Dual-path separates the cases: shown seals restore end-to-end accuracy; novel seals are a negative control (expected gold near zero). The scored output is the answer node or UNKNOWN, not (r_1, r_2) . We infer whether selection was supported from those outcomes and from controls (unique topology, written hop, matched vs. novel seals). Throughout, “selection failure” names that inference about the latent (r_1, r_2) stage; we do not observe the pair. Models may abstain (UNKNOWN) rather than guess (selective prediction; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). Coverage is the fraction of items with a non-UNKNOWN parse (other and both-named count; empty completions stored as UNKNOWN do not; NO_OUTPUT does not); selective accuracy is exact match among those items, not a forced-choice among the two routes (El-Yaniv and Wiener, 2010; Wen et al., 2025). NO_OUTPUT is an empty completion on a runner that does not store empties as UNKNOWN (Wiki-H5/Wiki-OO). **Both-named** is a parse that emits both person tails.

After sealing, the Eisner unique-path CONTEXT is Michael_Eisner | E334675310f05 | The_Walt_Disney_Company then The_Walt_Disney_Company | Eac04fab538ca | Burbank (12-hex tokens; entities remain English on the primary 2×2).

We use five measurement arms. **Readable English** is the baseline (start in q ; names and predicates unsealed). **Unique opaque** has one 2-hop and no plan: execution when topology uniquely determines the route. **Two-path opaque** has two same-type 2-hops and no plan: end-to-end behavior when two complete routes compete and the prompt supplies no semantic-to-symbol map. **Sealed join plan** writes (r_1, r_2) over σ -atoms (execution control); **SealRouter** is the non-LLM exact joiner on that sequence. **Free sealed English** seals the question and graph with no plan and no legend. On CEO, the question is hashed piecewise, so the full person atom in G typically does not appear in q (confounded; not the matched factorial). Additional scaffolds (name/key legends, sealed demonstrations) appear as ablations.

4 Method

Protocol terms. Headline WIKI-H5 quizzes are **isolation** files (one item per file; gold stored outside the prompt). Batch $n=12/n=24$ three-arm files use one shared key (Appendix A). A **lock** is a scored result file we do not overwrite. A **freeze** is a fixed record list. An **ARM** line is a one-line header that names the condition. **Named-arm** packing uses readable case ids plus that header; **hashed-id** packing keeps the ARM line and hashes the case ids; **no-ARM** packing drops the header. Unless noted, OpenAI uses 512 completion tokens with `reasoning_effort` omitted; **4096/medium** is 4096 completion tokens and `reasoning_effort` medium. Alias $k=3$ is three draws of the unpinned snapshot; the English-entity ambiguity sweep $k=1.5$ is a different k (Appendix B). Run tags and lock filenames are Appendix A. WIKI-H5 is the relation-only 2×2 ; header H5 keeps that ARM line.

Seal recipe. Atoms are normalized then HMAC-SHA256-renamed with a per-item 16-byte key (token E plus 12 hex digits). HMAC is keyed renaming, not a security mechanism; 48-bit truncation is instance-wise injective on these small atom sets (normalization rules, type non-separation, and collision checks: Appendix A). The intervention removes lexical semantics and also changes token-level surface statistics; permutation and alphanumeric generators assess that latter variation (Table 9). Hashing q token-by-token does not reconstruct the graph atom Michael_Eisner; that is the missing-start failure. Reported opaque cells are this HMAC construction, not every opaque identifier distribution.

Matched 2×2 (start in q). The $n=32$ suite is a seeded sample of 32 of 73 Wikidata-derived CEO records (Appendix A; Vrandečić and Krötzsch, 2014). Each instance is 4 triples (unique-path: 6

entities with a disconnected decoy island; two-path: 5 entities sharing the start). The gold two-hop uses `works_at` then the headquarters relation; the question says “led by”. The two-path condition *injects* a competing route: record $i+1$ ’s company/HQ under `partner_of/located_in`. That decoy is not a Wikidata fact. It isolates same-type competition on a fixed template, not naturally occurring CWQ/WebQSP ambiguity. To isolate relation selection from start-entity identification we keep the start string in the question and hold hop length, template, and triple count fixed (4 triples; unique-path has 6 entities, two-path 5). The intended 2×2 factors are relation labels (English vs. HMAC) and topology (one 2-hop from the start vs. two). Named-arm cells also change the ARM line and a readable case id with the condition (lines below); hashed-id and no-ARM packing are Section 5.2. Tails are intended as the same type; Wikidata P159 (headquarters location) was not filtered (buildings, stations, a street, and one state among gold HQs). Both OpenAI opaque two-path decoys sit on non-city golds with a city decoy tail (Appendix A). On Wikidata two-path graphs both intermediates are companies (`works_at/headquartered_in` vs. `partner_of/located_in`). In the primary run the gold 2-hop is listed before the decoy 2-hop in CONTEXT (same order in English and opaque). Listing-order robustness is Section 5.2: a new draw on the $n=200$ freeze, not a reordering of the primary completions. The $n=32$ shuffle is a later snapshot, not that comparison (Appendix A).

The primary two-path prompt is a single user message:

```
MODEL UNDER TEST. Read ONLY this file. Use CONTEXT only.
No other files.
Format: ANSWER_PLAIN[...]: <city_or_UNKNOWN>
ARM OPAQUE_AMBIG: opaque relations, English entities;
two 2-hops from start.
```

Named-arm ARM lines: ENG_UNIQUE one 2-hop; ENG_AMBIG two 2-hops; OPAQUE_UNIQUE opaque one 2-hop; OPAQUE_AMBIG two 2-hops (quoted); OPAQUE_AMBIG_PLAN same graph plus the sealed sequence. OE/OO Format is ANSWER_SEALED; ARM lines, verbatim from the isolation quizzes (same 32 graphs; start seal in q):

```
ARM OE_UNIQUE: opaque entities, English relations; one
2-hop.
ARM OE_AMBIG: opaque entities, English relations; two
2-hops.
ARM OO_UNIQUE: opaque entities and relations; one 2-hop;
start seal in q.
ARM OO_AMBIG: opaque entities and relations; two 2-hops;
start seal in q.
```

$n=200$ named-arm ids are `F200_{arm}_i`; $n=200$ hashed-id ids are `H200_` plus 10 hex; $n=32$ hashed-id ids use prefix `HDA_` (Section 5.2). That primary prompt does *not* contain the equality-recipe line (Appendix J). The format token is `<city_or_UNKNOWN>`, but six $n=32$ gold HQs are not cities (Appendix A). A model that obeys the format on those items must emit UNKNOWN. The powered city-typed cell is $n=200$ (Section 5.1); the $n=32$ lock is the three-family pin. OpenAI English two-path is 31/32 under that CONTEXT-only instruction, so the line does not by itself force abstention. When two complete same-type 2-hops compete and relation tokens do not map the question onto one route, the same instruction makes UNKNOWN the appropriate reply: CONTEXT gives no explicit cue for one city. Readable names (Eisner, Burbank) still leave OpenAI at UNKNOWN 24/32 on opaque two-path: either parametric HQ is treated as inadmissible, or the verb does not map onto one seal. We cannot separate those readings here (Section 7). The CEO three-arm decoy 2-hop begins at a different person; it is not the matched two-path factorial.

Further constructions. The relation-only factorial keeps entities English. A second isolation suite on the same 32 CEO records seals entities, relations, or both, always with the start token in q . Opaque-entity / English-relation (OE) is the ARoG-like cell: readable predicates, anonymous nodes. Both-opaque (Wiki-OO) seals entities *and* relations; it is a complementary ablation, not the primary relation-only 2×2 . Readable entity names therefore leave residual pretrained world knowledge as a possible cue; both-opaque is not a substitute for synthetic identifiers.

WikiMovies facts are sampled from the public MetaQA movie KB (Miller et al., 2016; Zhang et al., 2018). MOVIE-A32 and MOVIE-A100 leave English verbs such as *directed* in q ; MOVIE-B hashes question atoms except the bracketed start entity. We exclude a genre-vs-language variant because object type would bypass relation selection. This is not an evaluation on the official MetaQA 14K test split. Dual-path holds G fixed with two complete 2-hop routes and varies whether demonstrations expose the gold seals (Appendix D).

Scoring. The format asks for one answer token in the graph’s namespace or UNKNOWN (plain-text city when entities stay English; $\sigma(a)$ when entities are sealed). Primary metric: exact match (UNKNOWN is a miss). On opaque two-path that rate is how often the model commits to the gold city when CONTEXT gives no explicit cue for one city; it is not a score of following the CONTEXT-only instruction. The named-arm claim is the response mode (UNKNOWN vs. decoy). Scored UNKNOWN on Wiki-H5/Wiki-OO is an explicit UNKNOWN line. Dual and some no-hop OpenAI cells stored empty completions as UNKNOWN. On two-path arms we also report decoy and UNKNOWN rates (Table 3). Wilson 95% intervals use $z=1.96$ (Wilson, 1927) ($0/32 \Rightarrow [0.00, 0.11]$; $6/32 \Rightarrow [0.09, 0.35]$; $32/32 \Rightarrow [0.89, 1.00]$). Those intervals treat the 32 items as Bernoulli trials on a single draw; they are not a generation-variance interval. OpenAI WIKI-H5 $k=3$ at $n=32$ and $n=200$ is Section 5.1. McNemar p -values below are unadjusted across the reported contrasts. On the OpenAI Wikidata 2×2 , English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar exact two-sided complete-case $p=5.96 \times 10^{-8}$; McNemar, 1947). That is the two-path paired contrast, not an interaction test, and it largely measures removing the cue that made one sealed route identifiable. The empirical content is the response mode (UNKNOWN vs. decoy) and prompt-surface packing (Section 5.2). Isolation $n=32$ uses one quiz per file and a per-item HMAC. Cells marked **n.r.** were not evaluated; we never impute them as zeros.

Models. The primary family is OpenAI `gpt-5.6-sol` (Chat Completions; unpinned alias; temperature omitted; 512 completion tokens unless noted). Composer 2.5 and Grok 4.5 used Cursor’s agent SDK on the same quiz files, with a reply-only preamble OpenAI never received. Unique-path opaque still executes on those SDK cells (Composer 27/32, OpenAI 32/32, Grok 30/32; Table 8). Tools, retry, and empty-reply handling are not equated. The August lock stores the parsed answer, not the API `model` string. Inference settings and $k=3$ alias draws are Appendix A.

5 Results

Table 2 names the cells. OpenAI is primary; Composer/Grok used a different harness (Table 8). MOVIE-B is a cross-dataset check, not a Wiki-H5 resample. Packaging probes, plus Spider $n=100$ and 2Wiki-CF $n=200$, are Appendix G. JSON no-hop binding cells are Appendix E.

ID	Dataset	Condition	n	Models	Scored
WIKI-H5	Wikidata CEO	rel-only 2×2 , start in q	32	OpenAI (primary)	node
WIKI-H5-200	Wikidata CEO, city P31 (instance of)	same 2×2 , QID freeze	200	OpenAI	node
Header $\times$ decoy	same 32 people	ARM line $\times$ decoy, hashed ids	32	3 families	node
Header $\times$ decoy-200	same 200 people	ARM $\times$ decoy, hashed ids	200	OpenAI	node
Unique-no-ARM	same people	unique path, hashed ids, no ARM	32, 200	OpenAI	node
WIKI-OO	same graphs	both-opaque	32	3 families	$\sigma(a)$
WIKI-OO-200	$n=200$ freeze	both-opaque	200	OpenAI	$\sigma(a)$
WIKI-SHUF	same two-path	listing order	32	OpenAI	city
WIKI-SHUF-200	$n=200$ freeze	listing order	200	OpenAI	city
MOVIE-A32	WikiMovies	actor $\rightarrow$ movie $\rightarrow$ dir vs writer	32	3 models	person
MOVIE-A100	WikiMovies	same hops; verbs in q	100	OpenAI; Comp./Grok A	person
MOVIE-A200	WikiMovies	$n=100$ prefix + 100 new actors	200	OpenAI	person
MOVIE-B	WikiMovies	hashed verbs + two-hop ARM	100	OpenAI	person
MOVIE-B200	WikiMovies	Condition B on $n=200$ freeze	200	OpenAI	person
DUAL	WikiMovies/synth	matched vs. novel demos	32	3 models	$\sigma(a)$
DUAL-200	WikiMovies $n=200$	matched vs. novel	200	OpenAI	$\sigma(a)$
CEO-NL	Wikidata CEO	piecewise q , no plan	32	3 models	$\sigma(a)$
CEO-NL-200	$n=200$ freeze	piecewise q , no plan	200	OpenAI	$\sigma(a)$

Table 2: Cell inventory. WIKI-H5-200 is the powered city-typed freeze (OpenAI). Composer/Grok $n=32$ SDK cells: Table 8. Six $n=32$ P159 golds are not cities (Appendix A). A100 Condition A uses a thinner ARM than A32. Isolation cells use per-item HMAC.

5.1 Opacity $\times$ path ambiguity

WIKI-H5 holds the start string in q . The intended factors are relation lexicality (English vs. HMAC) and the number of 2-hops from that start (one vs. two). Named-arm packing (ARM line and readable case ids) also changes with the condition; hashed-id and no-ARM cells are Section 5.2.

City-typed $n=200$ (OpenAI). WIKI-H5-200 is a city-typed freeze of 200 people and companies (Appendix A); it does not overwrite the $n=32$ lock. Named-arm opaque two-path is 38/200 gold, UNKNOWN 151, decoy 11 (Table 3; Wilson [0.14, 0.25]); plan 200/200. The usual named-arm reply is UNKNOWN, not the decoy ($\approx 14:1$ vs. decoy). That mode is the named-arm headline. The 38/200 gold rate sits inside the $n=32$ Wilson band for 6/32. Decoy 0 on the $n=26$ city slice was a small-sample artifact: $n=26$ cannot resolve a $\sim 5\%$ decoy rate.

$n=32$ named-arm lock. On unique sealed routes OpenAI executes (opaque unique 32/32; plan 32/32). On competing sealed routes the lock is opaque two-path 6/32 (UNKNOWN 24, decoy 2); both decoys sit on non-city P159 golds. On 26 city-valued items unique English, unique opaque, and English two-path are 26/26; August city-slice opaque two-path is 5/26 (UNKNOWN 21, decoy 0, selective accuracy 5/5). Wilson: 6/32 $\Rightarrow$ [0.09, 0.35]; city slice 5/26 $\Rightarrow$ [0.09, 0.38]. The left panel of Figure 1 is this $n=32$ lock; the middle panel is $n=200$ named-arm; the right panel is hashed-id packing (Section 5.2). English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar $p=5.96 \times 10^{-8}$; McNemar, 1947). Opacity is not a randomized within-item treatment (Appendix I). Composer/Grok named-arm cells are Table 8 (English two-path is family-dependent, not a ranking). Alias $k=3$ does not replace these headline locks (Table 3; Appendix A).

OpenAI Wiki-H5 (gold / decoy / UNK)	Named $n=32$	Named $n=200$	Hashed $n=200$
English unique	31/0/1	200/0/0	—
English two-path	31/0/1	199/0/1	—
Opaque unique	32/0/0	200/0/0	—
Opaque two-path	6/2/24	38/11/151	134/20/44

Table 3: OpenAI response modes. Named-arm uses readable case ids. Hashed-id 512 keeps the ARM line (remainder: other 1, NO_OUTPUT 1). Coverage 49/200 $\rightarrow$ 155/200; selective accuracy 38/49 $\rightarrow$ 134/155. Alias $k=3$ gold 38/46/43 vs. 134/130/127 (gaps 96, 84, 84); coverage 49/56/58 vs. 155/153/145. Same-date $n=32$ hashed-id is 20/1/8 (other 2, NO_OUTPUT 1) at 512 tokens; Table 10’s H5-cyclic 20/32 is 4096/medium, a different run. Gold on all three $k=3$ draws is named-arm 28, hashed-id 97. Composer/Grok SDK cells: Table 8.

5.2 Prompt surface as a factor

The named-arm WIKI-H5 prompt contains this line:

```
ARM OPAQUE_ambiguous: opaque relations, English entities;
two 2-hops from start.
```

That sentence tells the model in English that two 2-hops exist. It is an uncontrolled treatment if abstention is attributed only to opacity-meets-ambiguity. Two manipulations—ARM line and case-id wording—must not be collapsed.

Case-id wording, ARM line held fixed. The case id appears in Format: ANSWER_PLAIN[id] and in the ID line. Named-arm uses readable ids (F22_OPAQUE_ambiguous_i); the H5-cyclic arm substitutes hashes. ARM line and CONTEXT are identical. On the city-typed $n=200$ freeze, hashed case ids at 512 tokens are Table 3 (UNKNOWN 151 $\rightarrow$ 44; gold +96; hashed-id has one NO_OUTPUT). Coverage and selective accuracy in that table show the extra commits are not only decoys. Under the CONTEXT-only instruction that is less abstention, not a better join: gold still relies on recall or residual cues (Section 3). Decoy also rises 11 $\rightarrow$ 20. Both prompts keep the ARM line, so this isolates case-id wording, not whether the prompt announces two 2-hops: hashing the id removes OPAQUE_ambiguous from Format/ID; the ARM line remains. The $n=32$ 512-token pair is Appendix A.

ARM line, case ids held hashed. Cyclic decoys attach record $i+1$'s company/HQ, as in WIKI-H5. Constant decoys reuse a pool record that does not share company, HQ, or person with the gold. Table 4 crosses the ARM line with cyclic vs. constant decoys at 4096/medium on the $n=200$ freeze. H5 cyclic vs. constant McNemar 95 vs. 10, $p=1.6\times 10^{-18}$; no-ARM 30 vs. 7, $p=1.9\times 10^{-4}$. The $n=32$ header $\times$ decoy cell (Table 10) already showed a simple-effect gap under H5 (cyclic 20 vs. constant 10; McNemar 14 vs. 4, $p=0.031$) and not under no-ARM (gold 25 vs. 27; McNemar 1 vs. 3, $p=0.625$). At $n=200$ both H5 and no-ARM cyclic-vs-constant simple effects are significant; we do not report an item-level interaction test. Of the 95 items that are gold under H5-cyclic and not under H5-constant, 74 become UNKNOWN and 21 take the constant decoy (McNemar 95 vs. 10). At $n=32$, all 14 cyclic-only golds become UNKNOWN (none take Cogent $\rightarrow$ DC). The drop is mostly abstention, not selection of the constant decoy's HQ. Isolation files show one item, so the model cannot see that the decoy repeats across the freeze. We still do not identify a mechanism. K2 is a shorter ARM that still names two hops:

ARM K2: opaque relations; 2 same-type 2-hop(s) from start.

K2 cyclic is near no-ARM cyclic (Table 4), so announcing two 2-hops is not what produces the H5 drop. ENG_AMBIG and OE_AMBIG carry that two-hop label and still score at ceiling (OpenAI 31/32 and 32/32; $n=200$ English two-path 199/200). Which H5 tokens matter (condition label vs. "English entities") is not identified here. Two-path K2 and no-ARM were not run at 512.

Listing order. Within one shuffle run on the $n=200$ freeze, gold-first is 25/100 and decoy-first is 34/100 (two-proportion $p\approx 0.16$, not significant). That is the listing test: same draw, same case-id family. Overall 59/200 versus the primary named-arm cell (Table 3) is a new draw, not a within-run reordering effect; we do not read +21 as a listing-order gain. The $n=32$ shuffle is a later snapshot (Appendix A): opaque two-path 5/32 (gold-first 3/16, decoy-first 2/16); decoy 0/16 vs. 1/16.

ARM line (OpenAI, hashed ids, 4096/medium, $n=200$)	Cyclic	Constant
no ARM	174/200	151/200
H5	129/200	44/200
K2	179/200	144/200

Table 4: Header $\times$ decoy on the city-typed $n=200$ freeze (same people as WIKI-H5-200). Cyclic = record $i+1$'s HQ; constant = `curve_decoy` ($n=200$: San_Jose $\times$ 199, London $\times$ 1; not Cogent $\rightarrow$ DC). English, no ARM, cyclic is 197/200. Unique-path was not run at 4096/medium in this packaging, so the no-ARM English vs. opaque pair is not an opacity $\times$ topology interaction. The 512-token hashed-id H5-cyclic cell is 134/200 (one NO_OUTPUT), a different budget from H5 cyclic 129/200 here. Cyclic gold=decoy is 0 at $n=32$ and $n=200$. $n=32$ golds: Table 10.

English Grok. Named-arm English two-path is Grok 4/32 (UNKNOWN 28). The English header $\times$ decoy cell (ENG_NONE_CYC) is 24/32. That cell is English relations, no ARM line, and hashed ids, so two things differ from the 4/32 named-arm English prompt. A prompt-verbatim Grok rerun of the named-arm English files stays 4/32, so the Cursor preamble is not that gap.

Exact-match under competing sealed routes moves with prompt surface. Case-id wording and the ARM line are large; a shared cause, if any, is not identified here. The 512-token hashed-id H5-cyclic cell is 134/200; Table 4 H5 cyclic is 129/200 at 4096/medium. Budgets differ, so these are not a ranking of manipulations.

Composer/Grok hashed-id H5-cyclic cells used `--preamble none` and score 6/32 and 0/32; the named-arm locks used the Cursor preamble and score 10/32 and 1/32. Both families go *down* with hashed ids, a sign reversal relative to OpenAI. We did not re-query Composer named-arm with the prompt file verbatim, so that reversal mixes case-id wording with preamble deletion and is not interpretable. Grok is at floor on both opaque two-path cells.

5.3 Both-opaque and readable relations

Table 5 is a second isolation suite on the same 32 records (ANSWER_SEALED; start seal in q). Unique-path both-opaque is 32/32 for all three families. Composer/Grok relation-only unique

Arm	Ent.	Rel.	OpenAI	Composer	Grok
Unique path	opaque	Eng.	32/32	32/32	32/32
Two paths	opaque	Eng.	32/32	28/32	3/32
Unique path	opaque	opaque	32/32	32/32	32/32
Two paths	opaque	opaque	1/32	0/32	0/32
Two paths + plan	opaque	opaque	32/32	28/32	32/32

Table 5: WIKI-OO and opaque-entity (OE) on the same 32 graphs (start in q). OpenAI OE two-path 32/32; both-opaque two-path 1/32 (UNKNOWN 30, decoy 1). Composer/Grok both-opaque two-path are UNKNOWN 32/32. Grok OE two-path 3/32 (UNKNOWN 29). Plan is a control. OpenAI $n=200$ both-opaque two-path is 12/200 (UNKNOWN 188, decoy 0).

(27/32, 30/32 in Table 8) is a different file, not evidence that hiding entities helps. Two-path both-opaque is OpenAI 1/32 (UNKNOWN 30, decoy 1; Wilson [0.01, 0.16]), Composer 0/32 (UNKNOWN 32), Grok 0/32 (UNKNOWN 32). A written hop recovers (32/32, 28/32, 32/32). OpenAI OE two-path (opaque entities, English relations) is 32/32: readable predicates restore end-to-end accuracy with anonymous nodes under this construction. Grok OE two-path remains 3/32 (UNKNOWN 29, decoy 0). On the $n=200$ freeze (OpenAI), unique both-opaque and OE two-path are 200/200; both-opaque two-path is 12/200 (UNKNOWN 188, decoy 0).

5.4 WikiMovies: hashed verbs and two-hop ARM

Table 6 uses the same actor→movie→director vs. writer template as MOVIE-A32 (person-valued tails). Hashed unique-path items still return the gold person. Condition A leaves *directed/starred* in q ; we do not treat those two-path scores as a selection result. Director is always the first person-tail, so parametric knowledge and a listing-slot heuristic are not separated. Condition B hashes those verbs and uses a two-hop ARM (Wiki-H5-style); two-path is mostly UNKNOWN (Table 6). A100 Condition A uses a thinner ARM (OPAQUE_AMBIG $n=100$); MOVIE-B’s ARM names two 2-hops. The $n=200$ Condition A cell uses the Wiki-H5 topology ARM, not that banner, so it is not a resample of 96/100. Condition B $n=100$ is OpenAI-only. The contrast is two factors on Wiki-Movies, not a same-prompt Wiki-H5 replication (Section 5.2). The $n=32$ MetaQA-style template cell (Appendix H) is not MOVIE-B.

WikiMovies people	Unique	Two-path
A32 opaque, verbs in q (OpenAI)	32/32	17/32
A100 Condition A, English, verbs in q	100/100	100/100
A100 Condition A, opaque, verbs in q	100/100	96/100
Condition B, hashed verbs + two-hop ARM	100/100	3/100
Condition B + plan	—	100/100
$n=200$ Condition A	—	122/200
$n=200$ Condition B	—	7/200
Composer / Grok Condition A two-path	—	62/100, 36/100

Table 6: MOVIE-A100/MOVIE-B: same actor→movie→director vs. writer template as MOVIE-A32 (Miller et al., 2016; Zhang et al., 2018) (start in q). Locked Condition B two-path 3/100 is UNKNOWN 87, decoy 0, both-named 10 (coverage 13/100, selective accuracy 3/13; Wilson [0.01, 0.08]); an archived ARM banner is 25/100. A32 unique is 32/32 on three families; Composer/Grok opaque two-path 17/32, 0/32; plan 32/32. On 32 A32 items matched by gold name (31 unique names) OpenAI is 17/32 vs. 29/32 under A100’s thinner ARM. A100 items 32–99 are 67/68 (index tail). A32 maps onto A100 indices 0–30 plus 41 (Gene Tierney is duplicated in A32), so index 41 sits in both counts and must not be added to 29/32. Keys differ; they cannot explain an English-name lookup.

5.5 In-prompt seal reuse

Dual-path holds G fixed with two complete 2-hop routes (Appendix D; Figure 2). Matched-relation reuses shown seals; it is not induction of a general cipher. Novel-relation demos teach seals that do not appear on the gold route (expected gold near zero). An explicit join plan is the execution upper

bound. Table 7 is OpenAI synthetic $n=32$, WikiMovies Dual $n=32$ (all three families), and the OpenAI WikiMovies Dual $n=200$ freeze, not the CEO freeze.

On a two-path graph a forced uniform guess among the two complete routes is $1/2$; that null does not apply once UNKNOWN is a third action. The pattern is consistent with prompt-local reuse of demonstrated seals and provides no evidence of systematic transfer to unseen relation tokens. We do not interpret this as a variable-binding mechanism.

Dual-path	Matched	Novel	Plan
OpenAI synthetic $n=32$	32/32	0/32	32/32
WikiMovies Dual $n=32$ (three families)	32/32	0/32	32/32
OpenAI WikiMovies Dual $n=200$	200/200	2/200	200/200

Table 7: DUAL: same G , two complete 2-hops (Appendix D; Figure 2 is OpenAI synthetic $n=32$). OpenAI matched is 32/32 under both noise regimes (decoy 0). Novel balanced is 0/32: all 32 stored UNKNOWN (24 empty completions, 8 explicit UNKNOWN). Novel asymmetric is 0/32 (UNKNOWN 24, of which 17 empty, decoy 8). WikiMovies Dual 0/32 is exact match; Composer novel decoy is 2/32. On the synthetic suite, Composer isolation-tag novel-relation UNKNOWN is a sealed-demo helper; Composer free-form is decoy 32/32. Grok free-form novel-relation asymmetric is UNKNOWN on all 32 items. Last-demo answer-token copy is 0/32 on every arm. OpenAI $n=200$ novel is UNKNOWN 198 explicit, decoy 0.

5.6 Missing-start control

The CEO isolation three-arm is an execution upper bound plus a missing-start cell, not the isolation of selection. Join plans score 32/32 on both plaintext and sealed atoms; SealRouter given the gold sealed relation sequence is 32/32; free sealed English scores 0/32 (all UNKNOWN) for Composer 2.5, OpenAI, and Grok 4.5 (Table 12). On the $n=200$ freeze, plan remains 200/200 and free sealed English 0/200. We exclude a batched free-sealed cell that can succeed by returning a unique neighbor without using q .

6 Discussion

Sealed unique routes are executable (Table 3; hashed-id no-ARM unique 200/200 at 512 tokens). Under two competing sealed routes, named-arm OpenAI usually abstains (Table 3). Hashed case ids on the same graphs reduce abstention (UNKNOWN 151 $\rightarrow$ 44 at 512 tokens, ARM line kept) and raise gold; that is packaging moving the commit rate, not a better CONTEXT join. Dropping the ARM line at 4096/medium is Table 4 (McNemar on the cyclic English/opaque pair; no-ARM cyclic opaque UNKNOWN 5). That no-ARM English/opaque pair is two-path at 4096/medium; unique-path was not run at that budget, so it is not an opacity $\times$ topology interaction. Alias $k=3$ (Table 3) does not make the two 512-token cells meet. That surface move is large; it does not make abstention the wrong description of the *named-arm* cell. Grok no-ARM opaque still prefers UNKNOWN over the decoy (23 vs. 3 cyclic); among 15 commits gold is 7 (Wilson [0.25, 0.70]). OpenAI’s higher no-ARM gold rate is not a CONTEXT join (Section 5.2). English predicates restore OpenAI with anonymous nodes (32/32 vs. both-opaque 1/32); Grok OE two-path is 3/32 (UNKNOWN 29). Dual-path reuses shown seals (matched 200/200); the novel-seal arm is a negative control (2/200; expected gold near zero). A high English two-path rate does not imply the same join once relations are opaque. OIR withholds a retrieved-path ranking and a local plaintext decoder on the free arms; PrivGemo’s CWQ/WebQSP numbers keep those paths and answer locally (Tan et al., 2026). We do not claim mathematical unidentifiability. Hypothesis status: Table 11.

7 Limitations

OpenAI is the primary family. Headline $n=200$ is the 15 September named-arm lock and the 512-token hashed-id lock; later alias draws do not replace them (Table 3; Appendix A). $n=32$ $k=3$ triples are Appendix A. Composer/Grok stay at $n=32$ on a different harness (Table 8). The competing two-path route is injected, not a Wikidata fact; graphs are four triples, smaller than retrieved KG-RAG

subgraphs. The $n=26$ decoy-0 slice is underpowered for a $\sim 5\%$ decoy rate (Section 5.1). The $n=32$ format token makes six P159 golds ill-posed; $n=200$ does not use those golds. The $n=32$ suite is 32 of 73 records, one template, one seed; decoys are cyclic, so item-independence for Wilson intervals is slightly optimistic (McNemar uses paired items but still assumes independent pairs). $k=3$ at $n=32$ mixes the August lock with later alias draws (Appendix A). The two later OpenAI named-arm draws are both 3/32 and Grok both 0/32; Composer is 7/32 then 6/32; hashed-id is 21/32 then 22/32. August named-arm is the highest draw in all three families. This is not a generation-variance CI. English entities leave parametric cues. Named-arm opaque two-path keeps those names and is still mostly UNKNOWN (24/32, 151/200), not a dump of the headquarters city. We cannot separate a recalled HQ treated as inadmissible from a verb that does not map onto one seal. 2Wiki-CF Wikipedia leak is 75/200 and OpenAI overlap leak 50/200 (Appendix G). Both-opaque unique is 200/200 with names sealed. HMAC two-path rates vary by generator (6/32, 15/32, 12/32). Those runs are undated; keyed HMAC and random permutation both emit E+12hex tokens, so the 6 vs. 15 gap is not a token-shape contrast (Table 9). The paper’s central distinction between selection and traversal is inferred from the answer node rather than measured: we never score an emitted (r_1, r_2) . Named-arm OpenAI selective accuracy stays high; Grok no-ARM commits are 7 gold / 15 (Wilson [0.25, 0.70]; Section 5.2). Header $\times$ decoy at $n=200$ is Table 4; Table 10 is the $n=32$ header $\times$ decoy cell. Two-path K2 and no-ARM were not run at 512. Unique no-ARM at 512 is not a matched-budget contrast with two-path no-ARM at 4096/medium; the no-ARM English/opaque pair at that budget is a two-path effect, not an opacity $\times$ topology interaction. MOVIE-A32 vs. MOVIE-A100 (17/32 vs. 29/32) is a thinner ARM on WikiMovies; $n=200$ Condition A uses the Wiki-H5 topology ARM, not that banner. Dual novel OpenAI UNKNOWN includes empty completions. Forced-choice, scored (r_1, r_2) , and an HQ-city swap are not in this evaluation. The ARM line was not randomized independently of case-id wording on the original lock. Composer hashed-id H5-cyclic used `--preamble none`; the named-arm Composer lock used the Cursor preamble, so that sign reversal is uninterpretable.

Ethical considerations

Public Wikidata CEO records (CC0; Vrandečić and Krötzsch, 2014) and the WikiMovies/MetaQA graph as released (Miller et al., 2016; Zhang et al., 2018). Appendix cells use 2WikiMultihopQA (Apache-2.0; Ho et al., 2020), Spider (CC-BY-SA-4.0; Yu et al., 2018), and FinQA as released (Chen et al., 2021). No personally sensitive information. Replies are hash-pinned, not a training dump. OIR is a diagnostic, not a confidentiality mechanism (English names remain on the primary 2×2).

8 Conclusion

Unique-route sealed execution succeeds with the start in q (Table 3; hashed-id no-ARM unique 200/200 at 512 tokens; both-opaque unique 32/32 on three families, Composer/Grok on the Cursor SDK). On competing sealed routes, named-arm OpenAI usually abstains (UNKNOWN, not the decoy). Hashed case ids, ARM line kept, reduce abstention at 512 tokens (UNKNOWN 151 $\rightarrow$ 44; Table 3; $k=3$ gaps 84–96). Dropping the header is a two-path cell at 4096/medium (unique-path was not run at that budget; Table 4). Exact-match under two sealed routes is strongly sensitive to prompt surface: case-id wording at 512 tokens; H5 vs. no-ARM McNemar at 4096/medium is Table 4 (cyclic and constant). Grok no-ARM opaque still prefers UNKNOWN over the decoy (23 vs. 3 cyclic); among 15 commits gold is 7 (Wilson [0.25, 0.70]; Section 5.2).

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A Protocol, scoring, and models

Quizzes (runs/) and locks (results/) are in <https://github.com/pankajjnits/oir> (Python ≥ 3.10 ; hash-pin results/SHA256SUMS, not HEAD). OpenAI: gpt-5.6-sol, no tools, temperature omitted. Named-arm, 512-token hashed-id, and unique no-ARM sidecars store returned_model gpt-5.6-sol (vendor alias, not a dated snapshot pin). August $n=32$ stores the parsed answer, not model/usage. Named-arm and 512-token hashed-id omit reasoning_effort. Header $\times$ decoy at 4096 sets reasoning_effort medium. $k=3$ sidecars also store returned_model. Named-arm $n=200$ OPAQUE_ambiguous sidecars are finish_reason stop on 200/200; no-ARM cyclic OPQ_NONE_CYC likewise 200/200 stop; hashed-id 512 OPQ_H5_CYC is stop 199 and length 1 (the NO_OUTPUT item). August $n=32$ stores no usage. Composer/Grok: Cursor agent SDK and a reply-only preamble OpenAI never received. Cite Dual free-form as *ff. Empty Dual/no-hop completions were stored as UNKNOWN; Wiki-H5 and Wiki-OO were not. We never impute n.r. as zeros.

Wiki-H5 seed 20260821, 16-byte keys from oir-fact-2x2-iso-v1; 32 of 73 CEO rows; decoys $i+1 \bmod 32$ (Wilson slightly optimistic). Wiki-H5 $n=32$ isolation is 32×5 named arms; $n=32$ header $\times$ decoy is 224 queries per family; $n=200$ header $\times$ decoy at 4096/medium is 1400. Hashed-id unique-path no ARM (16 September) is 27/32 (26/26 city) and 200/200 (512 tokens; same CONTEXT as OPAQUE_UNIQUE); the five $n=32$ misses are non-city P159 format tokens that named-arm unique emitted. Wiki-OO uses oir-entrel-2x2-iso-v1; CEO-NL seed 20260820. The $n=12/n=24$ three-arm is **batch** (one shared key oir-ceiling-three-arm-v1); isolation $n=32$ is per-item keys. Wiki-H5-200 is a separate freeze (data/real/wikidata_ceo_hops_n200.json, seed 20260915; 200 unique people and companies; HQ frequency capped at 3; P31 $\in \{Q515$

city, Q1549591 big city, Q1093829 city of the United States, Q1637706 city with millions of inhabitants, Q3957 town, Q5119 capital, Q13539802 place with town rights and privileges, Q5153359 municipality of the Czech Republic, Q7930989 city or town}; person/company/HQ QIDs stored; all 200 P169 (chief executive officer)+P159+P31 triples re-checked live against the Query Service; SPARQL verify 200/200). Labels are Wikidata English labels: Vienna/The Hague/Mattersburg use Q13539802; Prague/Čáslav use Q5153359. HMAC oir-fact-2x2-iso-n200-v1; n=32 prompts are not rewritten. WikiMovies n=200 keeps the n=100 prefix and adds 100 actors.

Run-tag manifest. Body text names cells by packing and date. Tags:

Cell	Tag
Named-arm n=32 August	gpt56
Named-arm n=32 15 Sep	gpt56k3r2/r3
Named-arm n=32 11 Sep 512 (OPAQUE_AMBIG only)	gpt56sep
Hashed-id n=32 11 Sep 512	gpt56abl512
Hashed-id n=32 4096/medium	gpt56abl
Named-arm n=200 15 Sep	gpt56n200
Hashed-id n=200 512	gpt56n200h512
Header×decoy n=200 4096/medium	gpt56n200abl
n=200 k=3 16 Sep	*k3r2/*k3r3 on those two 512 stems
Unique no-ARM n=200 / n=32 512	gpt56n200uninone / gpt56uninone
n=32 shuffle	gpt56 on the shuffle stem
Grok named-arm English verbatim	grok45pn
Composer Dual isolation / free-form	composer25 / composer25ff
Grok Dual free-form	grok45ff

Lock JSON lives under results/ with that tag as the filename suffix (e.g. ...gpt56n200.json, ...gpt56n200h512.json). gpt56n200h512 queried only OPQ_H5_CYC. The n=200 shuffle stem is factorial_2x2_iso_shuffle_n200_ (case ids F200S_ rather than F200_). It is not a second model snapshot and does not overwrite the headline lock.

Non-city P159 golds: Broadway, NRK_Marienlyst, Fox_Plaza, CNN_Center, Lod_Railway_Station; New_Jersey is a state. OpenAI two-path decoys are items 27 and 31 (both non-city). City slice 5/26 (decoy 0) is underpowered; n=200 is decoy 11.

Alias k=3. Alias draws of the unpinned snapshot are not a generation-variance interval. Each packing's k=3 is that packing's headline lock plus two later alias draws of the same packing. At n=32: August named-arm lock 6/32 (Composer 10/32, Grok 1/32); 15 September alias draws 3/32, 3/32 (Composer 7, 6; Grok 0, 0). Named-arm k=3 is that August headline plus the two 15 September alias draws (6/3/3); the 11 September named-arm 5/32 is the same-date hashed-id pair, not a later alias of the August headline, so it is not in that triple. Hashed-id k=3 is 20/21/22 because the hashed-id headline is the 11 September lock 20/32, plus 15 September 21, 22 (512 tokens, reasoning_effort omitted). On 26 city-valued items named-arm opaque two-path is 5/26, 2/26, 2/26 across those three named-arm draws. On OpenAI the other n=32 named-arm cells hold: unique opaque 32/32, 31/32, 32/32; English two-path 31/32, 30/32, 30/32; plan 32/32 on every draw. Named-arm English two-path is OpenAI 31/32, Composer 26/32, Grok 4/32. Opaque unique stays high (Composer 27/32, Grok 30/32); a written hop recovers (28/32, 32/32). August named-arm is highest in all three families. At n=200 the headline pair and k=3 golds are Table 3. Those later draws do not replace the headline locks.

Case-id n=32 (512 tokens). The same-date pair is 11 September: 5/32 vs. 20/32 (+15; one NO_OUTPUT each; reasoning_effort omitted). Item 0 named-arm Format is ANSWER_PLAIN[F22_OPAQUE_AMBIG_0]; hashed-id H5 is ANSWER_PLAIN[HDA_4eba493d71]. Against the named-arm triple only +18 and +19 are same-day.

An archived Movie-B ARM banner is 25/100; locked topology-only B is 3/100. CEO free English hashes q piecewise.

Normalization and HMAC. Strip; collapse whitespace to `_`; map other non-alphanumeric characters (except underscore, period, hyphen) to `_` using `[A-Za-z0-9_.-]` (`Zürich`→`Z_rich`; pure CJK keeps Unicode); truncate to 64; empty maps to `EMPTY`. The CEO JSON already deleted diacritics upstream. Distinct raw atoms must normalize to distinct strings. HMAC-SHA256 is over UTF-8 bytes with a per-item 16-byte key; the token is E plus the first 6 digest bytes as 12 hex digits. Truncation is not injective over $\{0, 1\}^*$; 48-bit width is instance-wise injective on these small atom sets. Entity and relation types are not domain-separated. On each scored Wikidata 2×2 instance the normalized entity strings and relation strings are disjoint, and no HMAC collision among prompt atoms was observed. Sealing is deterministic given the instance key; the same plaintext need not map to the same token across prompts.

Composer/Grok named-arm ($n=32$). Table 8 is the SDK roster, not a ranking of families. UNKNOWN is an explicit line (missing/unparsed 0; both-named 0). Composer “other” is truncated gold Lod on `ENG_UNIQUE`, `OPAQUE_UNIQUE`, and `plan`, plus Los Angeles for gold `Fox_Plaza` on `OPAQUE_UNIQUE`.

Wikidata named-arm (iso. $n=32$)	OpenAI	Composer 2.5	Grok 4.5
English, unique	31/0/1/0	29/0/2/1	32/0/0/0
English, two paths	31/0/1/0	26/0/6/0	4/0/28/0
Opaque, unique	32/0/0/0	27/0/3/2	30/0/2/0
Opaque, two paths	6/2/24/0	10/0/22/0	1/0/31/0
Opaque two paths + plan	32/0/0/0	28/0/3/1	32/0/0/0

Table 8: Four-way outcomes: gold / decoy / UNKNOWN / other. Composer 2.5 and Grok 4.5 used Cursor’s agent SDK with a reply-only preamble OpenAI never received. Body Table 3 is the OpenAI lock.

B Generator robustness and $n=32$ header×decoy

Opaque generator (OpenAI, start in q)	Unique	Two-path
Keyed HMAC (E+12hex)	32/32	6/32
Random permutation (E+12hex)	31/32	15/32
Random alphanumeric (not HMAC-shaped)	29/32	12/32

Table 9: Generator robustness on the same 32 Wikidata graphs and question template. Unique-path accuracy exceeds two-path accuracy under three independently generated opaque relation maps; the two-path *rate* is not invariant. Wilson: $29/32 \Rightarrow [0.76, 0.97]$, $12/32 \Rightarrow [0.23, 0.55]$, $15/32 \Rightarrow [0.31, 0.64]$. Alphanumeric two-path is all abstention or gold (decoy 0).

A random permutation of HMAC-shaped labels on the same graphs preserves unique > two-path (OpenAI 31/32 vs. 15/32; Composer 28/32 vs. 2/32; Grok 31/32 vs. 0/32). An English-entity ambiguity sweep $k=1..5$ (constant decoy) keeps OpenAI at 29, 29, 28, 27, 28; Composer falls to 11/32; Grok is 0/32 for $k \geq 2$. Disney names remain, so k is not isolated from entity semantics.

Header×decoy uses hashed case ids that do not name the condition on the same 32 people (`header_decoy_ablation_iso_*.json`; body Section 5.2). Cells are named by header and decoy: H5 keeps the Wiki-H5 ARM line, K2 keeps a shorter two-hop ARM, and “no ARM” deletes the line; cyclic decoys attach record $i+1$, constant decoys reuse the first pool record that does not share company, HQ, or person ($n=32$: Cogent Communications → Washington, DC, gold=decoy 0; $n=200$: San_Jose ×199, London ×1, gold=decoy 0). OpenAI 4096/medium $n=32$ golds:

Composer/Grok the same day, prompt file verbatim (`--preamble none`): H5 6/2 and 0/0; K2 20/13 and 0/0; no ARM 21/18 and 6/1; English 26 and 24. Named-arm English Grok verbatim (`grok45pn`) stays 4/32: the Cursor preamble is not that cell. `ENG_NONE_CYC` 24/32 differs in both ARM and case ids (Grok 24/32 vs. no-ARM opaque 6/32; McNemar 18 vs. 0). Grok no-ARM opaque is cyclic 6/3/23 (gold / decoy / UNKNOWN) and constant 1/5/26; among 15 commits, gold is 7 (Wilson $[0.25, 0.70]$). Constant decoy does not raise the Wiki-H5-header cell on any family.

ARM line (OpenAI, hashed ids, 4096/medium)	Cyclic	Constant
no ARM	25/32	27/32
H5	20/32	10/32
K2	30/32	26/32

Table 10: Header×decoy gold counts on the same 32 people. English, no ARM, cyclic is 29/32. H5 McNemar 14 vs. 4, $p=0.031$; no-ARM gold 25 vs. 27, McNemar 1 vs. 3, $p=0.625$. Body Table 4 is the $n=200$ lock. On that freeze, no-ARM cyclic English vs. opaque is McNemar 25 vs. 2 ($p=5.65\times 10^{-6}$). H5 vs. no-ARM on constant decoys is McNemar 4 vs. 111. H5 vs. no-ARM on cyclic decoys is McNemar 6 vs. 51 ($p=5.7\times 10^{-10}$). No-ARM cyclic opaque is gold 174, decoy 17, UNKNOWN 5, other 4.

C Hypothesis status

Hypothesis	Test	Status
H1. Opacity alone blocks traversal	unique-path, start in q	not supported (32/32)
H2. Written hop restores e2e (control)	two-path + plan; SealRouter	control (32/32)
H3. Input-binding control (CEO piecewise q)	CEO isolation NL	control (0/32); not selection
H4. Matched demos restore shown seals	dual-path matched vs. novel	OpenAI 32/32 vs. 0/32
OA. Named-arm two-path usually abstains (UNKNOWN, not the decoy) when opaque relations meet path ambiguity	matched 2×2, start in q , named-arm packing	$n=200$ UNKNOWN 151, decoy 11, gold 38/200; $n=32$ UNKNOWN 24, decoy 2, gold 6/32 (24 of 26 non-gold abstain); named-arm $n=32$ McNemar 25 vs. 0 (city 21 vs. 0); Sec. 5.1
H6. Movie KB, hashed verbs + two-hop ARM (two-factor check)	WikiMovies Cond. B $n=100$	OpenAI two-path 3/100; archived banner 25/100
H7. Readable relations with opaque entities	Wikidata OE vs. Wiki-OO	supported for OpenAI here
H8. Prompt surface moves two-path exact match on the same graphs	case-id at 512; two-path ARM vs. no-ARM at 4096/medium	$n=200$ case-id +96; $n=200$ no-ARM hashed cyclic English vs. opaque McNemar 25 vs. 2; H5 vs. no-ARM McNemar cyclic 6 vs. 51, constant 4 vs. 111; Sec. 5.2

Table 11: Hypothesis status. Numbers skip H5 so they do not collide with the Wiki-H5 ARM line in Section 5.2. H4 exact-match is OpenAI Dual; Composer free-form novel is decoy 32/32 (composer25ff). OA is named-arm Wiki-H5 (OpenAI); the usual reply is UNKNOWN, not the decoy. The exact-match drop consists mostly of abstentions (24 of 26 non-gold at $n=32$). H6 hashes verbs *and* uses a two-hop ARM; an archived banner is 25/100. H7 is OpenAI-specific (OE vs. Wiki-OO). H8 is two-path prompt surface on the same graphs (case-id +96 is Table 3; no-ARM hashed cyclic English vs. opaque McNemar and ARM-line McNemar are Table 4 and Table 10), not cyclic vs. constant (which changes CONTEXT).

In this table, row OA is the named-arm abstention hypothesis (named-arm Wiki-H5 response modes; McNemar is the paired exact-match contrast). The no-ARM hashed cyclic English vs. opaque contrast is H8: a two-path packing cell at 4096/medium, not an opacity×topology interaction. H5 in Section 5.2 is the Wiki-H5 ARM line, not a hypothesis number.

Table 12 is the missing-start CEO-NL control, not Wiki-H5.

CEO-NL isolation $n=32$	Composer 2.5	OpenAI	Grok 4.5
Sealed plan / SealRouter	32/32	32/32	32/32
Free sealed English (piecewise q)	0/32	0/32	0/32

Table 12: Missing-start control, not WIKI-H5. Isolation $n=32$. Batch $n=12/n=24$ uses one shared key (Appendix A).

D Dual-path arms

We name the arms by what the demos teach. **Asymmetric noise**: extra edges hang only off the gold midpoint (the trap path looks cleaner). **Balanced noise**: symmetric extra edges on both midpoints. **Matched-relation** (asymmetric or balanced): demos expose the gold relation seals; the decoy path uses different seals. **Novel-relation** (asymmetric or balanced): demos teach seals that do not appear on the gold route; both routes are novel at quiz time. **Explicit join plan**: gold relation sequence is written in the prompt.

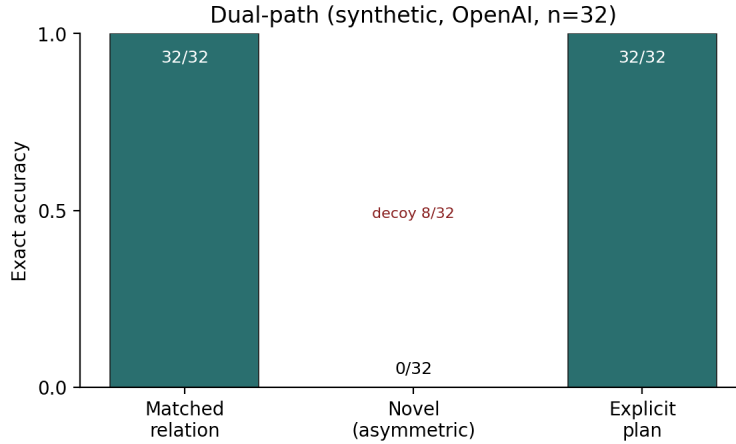


Figure 2: DUAL (synthetic, OpenAI $n=32$ isolation): matched 32/32 under both noise regimes (one bar; both 32/32); novel asymmetric 0/32 (decoy 8); explicit plan 32/32.

Isolation $n=32$ is the reported cell. Cite Dual free-form as `composer25ff/grok45ff`; the isolation-tag Composer helper is not free-form. Local `mistral/qwen35` are not in the roster (plan n.r.).

E No-hop binding (JSON)

Prose labels in Table 13 map to lock JSON arms: schema list LEGEND and English protocol META (`seal_layer_legend_small_n32_harness*.json`); silent ISO_SILENT, equality recipe ISO_RULE, and broken match NS_SILENT (`opaque_iso_json_n32_harness*.json`); two-route recipe TWO_REL and vague hints ABSTRACT (`opaque_hint_reason_n32_harness*.json`). Prompts are fixed English blocks, not WIKI-H5. Equality means token identity (Appendix J). English JSON keys alone do not restore the walk (1/6 on the $n=6$ batch; that arm is not run at $n=32$).

F Sealing middleware

Bind a typed full name to one graph atom, then seal. Isolation without binding is 0/6 vs. layer 6/6 (Composer, OpenAI, unversioned Claude; plaintext in the packed prompt: 0). Those files attach a join: an execution control, not the free WIKI-H5 6/32 cell.

Arm (no written hop)	OpenAI	Composer 2.5	Grok 4.5
Schema list only	0/32	0/32	0/32
English protocol (no hop)	32/32	32/32	32/32
Same σ , silent	6/32	19/32	31/32
Same σ + equality recipe	32/32	32/32	32/32
JSONKEY : match broken	0/32	0/32	0/32
Sealed Q, two routes, recipe	31/32	25/32	32/32
Sealed Q, two routes, vague hints*	13/32	23/32	31/32

Table 13: Selection support without a written hop (Wikidata CEO JSON; city or mentor country is the distractor; isolation; per-item HMAC). Schema-only and broken token overlap fail for all three. An English protocol or an equality recipe that does not name the field saturates. Silent same- σ overlap and vague hints are family-dependent, not a three-family restore. OpenAI silent 6/32 and vague-hint 13/32 include 24 and 15 empty completions stored as UNKNOWN. *Composer vague-hint distractor 1/32. JOIN was not evaluated at $n=32$. Wilson: 32/32 $\Rightarrow$ [0.89, 1.00], 0/32 $\Rightarrow$ [0.00, 0.11], 6/32 $\Rightarrow$ [0.09, 0.35], 31/32 $\Rightarrow$ [0.84, 0.99]. Silent same- σ OpenAI 6/32 is a JSON-binding cell, distinct from WIKI-H5. NS_SILENT ARM lines say “Match impossible”; those 0/32 rows are not a sealed-join collapse.

G Packaging probes

Not the matched 2×2 . Free sealed English 0/12 vs. plan 12/12 on table/JSON/vault (Composer, OpenAI). Spider identifier-in- q $n=100$: EQ 51/100, namespace-break 0/100 (OpenAI; not official EM). If the identifier is not in q ($n=26$), the namespace-break cell is 20/26: an English q plus legend can still bind. Excel NL names both predicates: Composer 0/16 vs. plan 16/16. FinQA NL 0/8 vs. gold programs 8/8 (Chen et al., 2021). 2Wiki-CF: Composer plain NL 200/200; OpenAI 150/200 with overlap leak 50/200 (not a 2Wiki eval). Long context: NL 0/8, plans 8/8. Rename under a second HMAC: 12/12. SealRouter and gold-SQL are non-LLM controls (Figure 3): gold-SQL is the Spider gold query executed after name sealing (191/200), not an exact join.

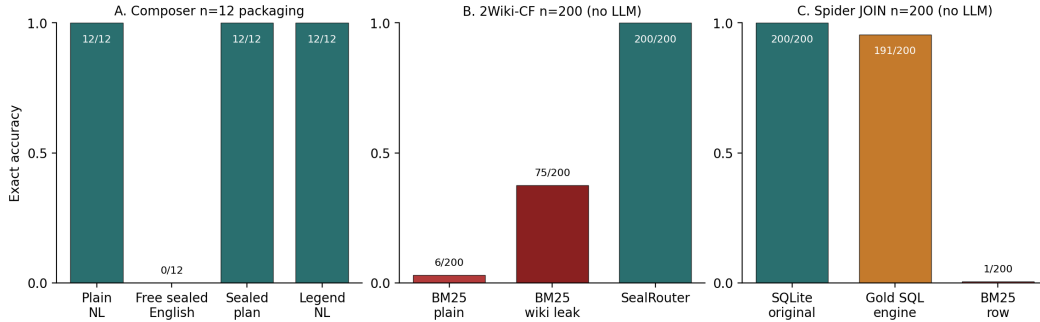


Figure 3: Non-LLM and packaging controls. Left: Composer $n=12$ (plain 12/12, free sealed English 0/12, plan 12/12, legend 12/12). Middle: 2Wiki-CF lexical BM25 6/200 (not a 2Wiki eval), Wikipedia leak 75/200, SealRouter 200/200. Right: Spider sqlite $n=200$ (200/200), gold-SQL 191/200 (sealed names; 9 execution misses), lexical BM25 1/200 (Robertson and Zaragoza, 2009). Spider identifier-in- q EQ 51/100 is a different cell.

H Extended scores

MetaQA-style 2-hop templates on the same public movie graph are a question-form control, not an evaluation on the official MetaQA 14K test split. Plaintext is 32/32 for all three families. Condition A (English question over opaque relations) is OpenAI 32/32, Composer 19/32, Grok 15/32. Condition B hashes those verbs except [start]: OpenAI 25/32, Composer 27/32, Grok 14/32 (plans 32/32, 31/32, 32/32). That $n=32$ template cell is not MOVIE-B.

I Sign-flip of the two-path opacity contrast

Let $d_i = \mathbf{1}[\text{opaque correct}] - \mathbf{1}[\text{English correct}]$ and $c_i = d_i^{\text{two}} - d_i^{\text{unique}}$. Of 32 WIKI-H5 items, 24 have $c_i = -1$, one has $c_i = -2$, 7 have $c_i = 0$; $T = -26$. Unique-path opaque is 32/32, so d^{unique} is nonzero on one item and the c_i mostly copy the two-path McNemar. The construction is an item-level interaction contrast whose null is exchangeability of unique-path and two-path opacity effects ($p = 2^{-24} = 5.96 \times 10^{-8}$). With unique-path at ceiling it cannot say anything beyond the two-path simple effect, so the identical p with McNemar 25 vs. 0 (McNemar, 1947) is expected. Opacity is not a randomized within-item treatment. The body reports Wilson intervals and the paired counts.

J Equality recipe (JSON cell)

Verbatim protocol used in Table 13, *not* in WIKI-H5: “QUESTION and CONTEXT use the same encoding. Identical tokens are the same entity or the same field (equality, not English). START is a QUESTION token that also appears as an id value in CONTEXT. The asked field is a QUESTION token that also appears as a JSON key. Walk CONTEXT by token equality. If more than one reading is possible, UNKNOWN.”

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